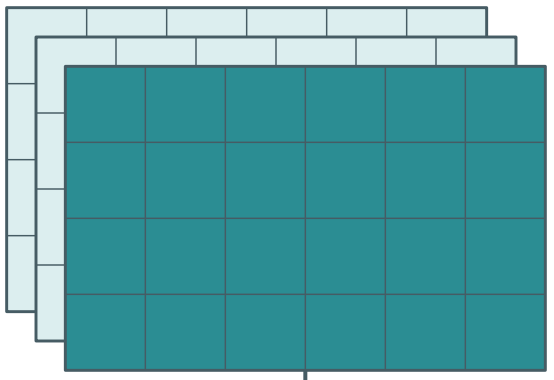
335 exposures

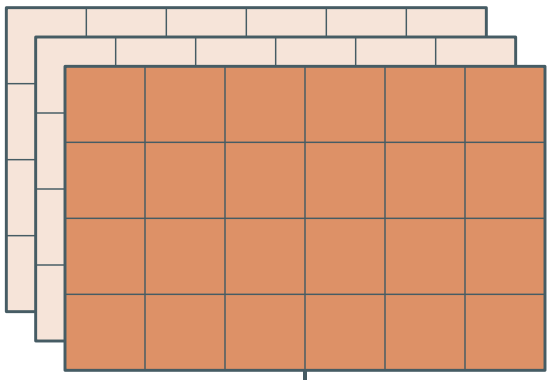
7 categories



18 annual surveys

149 phenotypes

9 categories



Eligible pairs

≥2 years; ≥500 complete cases

Merge and weight

Survey weights, strata and PSU

Weighted standardization

Continuous exposures: log10 transformation
Phenotypes: original scale

Environmental extension

2020–2021; single survey
19 exposures × 23 phenotypes
Separate weights; 437 tests

Covariates

Age, age², sex
Education, income, survey year
Energy / creatinine, as applicable

Survey-weighted regression

Covariate-adjusted linear models

Summary statistics

β, SE, P, R^2 and ΔR^2
Bonferroni and BY FDR

Robustness analyses

Annual recurrence; I^2
Age interactions
Covariate sensitivity

Vibration of effects

All eligible atlas pairs
12 factors; ≤64 adjustment sets
Direction / significance stability

Joint exposure models

≤20 BY-selected exposures
Imputation (m = 10); ΔR^2

KNHANES atlas

Association patterns and explained variance